⚙ AWS Compute

Module 2 — AWS Technical Essentials

💻 What Is Amazon EC2?

Resizable virtual machines launched in minutes. Pay only while running.

- Full OS-level control (Linux, Windows)
- Choose CPU, memory, storage, network
- Runs inside your VPC
- Install any software you need

🛠️ EC2 Instance Type Families

- **General Purpose (M, T)** — balanced workloads
- **Compute Optimized (C)** — CPU-intensive tasks
- **Memory Optimized (R, X)** — in-memory databases
- **Storage Optimized (I, D)** — high disk throughput

Naming: e.g. `m5.xlarge` = family (m) + generation (5) + size (xlarge)

💰 EC2 Pricing Models

On-Demand

- Pay by the second/hour
- No commitment required

Reserved / Savings Plans

- 1 or 3-year commitment
- Up to 72% discount

Spot Instances

- Up to 90% discount
- Can be interrupted by AWS

Dedicated Hosts

- Physical server for you
- Compliance / licensing needs

🔄 EC2 Instance Lifecycle

Launch → Running → Stop / Terminate

- **Running:** billed, fully operational
- **Stopped:** no compute charges, EBS persists
- **Terminated:** instance deleted permanently
- **Rebooted:** keeps public IP and data

📦 Containers on AWS

Amazon ECS

- AWS-native orchestrator
- Tight AWS integration
- Simpler to start

Amazon EKS

- Managed Kubernetes
- Open-source ecosystem
- Portable across clouds

AWS Fargate

- Serverless containers
- No EC2 to manage
- Works with ECS or EKS

⚡ AWS Lambda

Run code without provisioning servers. Pay only per invocation.

- Event-driven (S3, API Gateway, etc.)
- Auto-scales to demand
- Supports Python, Node.js, Java, Go, .NET
- Max 15 minutes per execution

🎯 Choosing the Right Compute

💻 EC2 When...

- Full OS control needed
- Long-running processes
- Legacy app migration

⚡ Lambda When...

- Event-driven, short tasks
- Variable / spiky traffic
- Zero admin overhead desired

Containers (ECS/EKS) fit between: microservices needing portability without full VM overhead.

🛠️ Lab Time

Launch the Employee Directory App on EC2

⏱️ 25 minutes

- Launch an EC2 instance with a web application
- Configure security group for HTTP access
- Verify the application in a browser